Automated Mutual Explicit Induction in Separation Logic

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Software and verification

Photos from the Internet and organized by Andreea Costea

Verification, Hoare logic, and Separation logic

Hoare logic: verifies functional correctness: $\{P\}$ S $\{Q\}$

Separation logic: reasons about imperative programs mutating data structures (C/C++/Java, etc).

- Assertions serve as pre-conditions and post-conditions (like Hoare logic): $\{P\}$ S $\{Q\}$
- Assertions model the disjointness of the computer memory (unlike Hoare logic)

$$x \mapsto 1 * y \mapsto 2$$

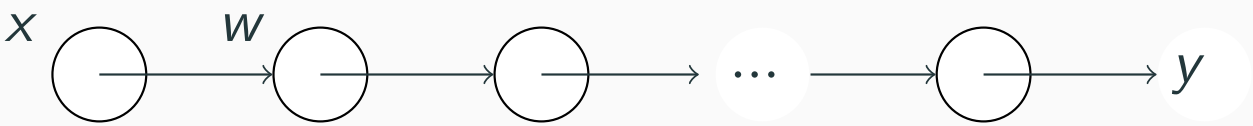
$$\{x \mapsto 1 * y \mapsto 2\} \ {}^*\mathrm{x} = 100 \ \{x \mapsto 100 * y \mapsto 2\}$$

Separation logic

Predicates modelling the computer memory:

- the empty memory: **emp**
- content of the memory: $\mathbf{x} \mapsto \mathbf{1}$
- disjointness of the memory: $\mathbf{x} \mapsto \mathbf{1} * \mathbf{y} \mapsto \mathbf{2}$
- inductive definition describing data structures:

$$\begin{aligned} \mathbf{ls(x,y)} \quad \triangleq \quad & \mathsf{emp} \wedge x = y \\ & \vee \ \exists w.(x \mapsto w * \mathsf{ls}(w,y)) \end{aligned}$$



Separation logic entailment

During verification of programs, the verifiers generate constraints in the form of separation logic entailments.

Example: $\mathbf{tmp(x) \vdash \exists y.\, ls(x, y)}$

with

$$
\begin{array}{lcl}
\mathsf{tmp}(x) & \triangleq & \mathsf{emp} \\
 & & \vee \;\; \exists u.(x \mapsto u * \mathsf{tmp}(u)) \\
 & & \vee \;\; \exists u, v.(x \mapsto u * u \mapsto v * \mathsf{tmp}(v)) \\
\\
\mathsf{ls}(x, y) & \triangleq & \mathsf{emp} \wedge x = y \\
 & & \vee \;\; \exists w.(x \mapsto w * \mathsf{ls}(w, y))
\end{array}
$$

Research topic and outline

Research topic:

How to prove separation logic entailments containing inductive heap predicates?

Outline:

- Backgrounds
- Our proposal: A mutual inductive proof system
- Experiment
- Conclusion and future works

Background: separation logic

Separation logic model: $s, h \models F$

Semantics of entailment:

$$F \vdash G \text{ is Valid iff } \forall s, h.\ (s, h \models F \rightarrow s, h \models G)$$

where

s : stack model, capturing the variables' value,

h : heap model, representing the memory's state.

Background: separation logic

Approaches to prove separation logic entailments:

- Automata-based, graph-based: decision procedures for restricted inductive heap predicates: [Iosif et al., 2013], [Piskac et al., 2013], [Enea et al., 2014], etc
- Proof theory: handle pre-defined predicates (automatically): [Berdine et al., 2004], or user-defined predicates (semi-automatically, using lemmas): [Chin et al., 2012],[Madhusudan et al., 2012], etc
- Induction-based proof theory: handle user-defined inductive predicates, automatically, but are incomplete techniques: [Brotherston et al., 2011], [Chu et al., 2015]

Background: mathematical induction

Given a conjecture $\mathcal{P}(\alpha)$:

- α is a structure of type τ,
- $\prec_\tau$ is a well-founded relation on τ

Well-founded induction (Noetherian induction):

In order to prove $\mathcal{P}$ holds for all α, it suffices to prove that:
$\mathcal{P}$ holds for all α, whenever $\mathcal{P}$ holds for all $\beta \prec_\tau \alpha$

Formally:

$$\frac{\forall \alpha : \tau.\ (\forall \beta : \tau.\ \beta \prec_\tau \alpha \to \mathcal{P}(\beta)) \to \mathcal{P}(\alpha))}{\forall \alpha : \tau.\ \mathcal{P}(\alpha)}$$

Induction proof using model order in separation logic

Model order:

- Entailment model: $s, h \models (F \vdash G)$ iff $s, h \models F \rightarrow s, h \models G$
- $s_1, h_1 \prec s_2, h_2$, iff $|h_1| < |h_2|$. ($\prec$ is a well-founded relation)

Induction principle:

In order to prove $F \vdash G$ holds for all model s, h (i.e. $F \vdash G$ is valid), it suffices to prove that: $F \vdash G$ holds for all model s, h, whenever $F \vdash G$ holds for all model $s', h' \prec s, h$

Formally:

$$\frac{\forall s, h.\ (\forall s', h'.\ s', h' \prec s, h \rightarrow s', h' \models (F \vdash G)) \rightarrow s, h \models (F \vdash G)}{\forall s, h.\ s, h \models (F \vdash G)}$$

Prove the motivating example: $\mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x, y)$

Denote $\mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x, y)$ as **E**, and

- record **E** as an induction hypothesis
- unfold $\mathsf{tmp}(x)$ in the left-hand side (LHS) of **E**

(base step)	(inductive step)	(inductive step)
(E_1)	(E_2)	(E_3)
$\mathsf{emp} \vdash \exists y.\, \mathsf{ls}(x, y)$	$x{\mapsto}u * \mathsf{tmp}(u) \vdash \exists y.\, \mathsf{ls}(x, y)$	$x{\mapsto}u * u{\mapsto}v * \mathsf{tmp}(v) \vdash \exists y.\, \mathsf{ls}(x, y)$

$$\frac{(E_1) \quad (E_2) \quad (E_3)}{(\mathbf{E})\ \mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x, y)}$$

with

$$\begin{aligned} \mathsf{tmp}(x) \triangleq\ & \mathsf{emp} \\ & \vee\ \exists u.(x{\mapsto}u * \mathsf{tmp}(u)) \\ & \vee\ \exists u, v.(x{\mapsto}u * u{\mapsto}v * \mathsf{tmp}(v)) \end{aligned} \qquad \begin{aligned} \mathsf{ls}(x, y) \triangleq\ & \mathsf{emp} \wedge x{=}y \\ & \vee\ \exists w.(x{\mapsto}w * \mathsf{ls}(w, y)) \end{aligned}$$

Inductive step: prove (E_2) $x{\mapsto}u * \mathsf{tmp}(u) \vdash \exists y.\, \mathsf{ls}(x, y)$

Denote $\mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x, y)$ as **E**, and

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Inductive step: prove $(E_2)\ x{\mapsto}u * \mathsf{tmp}(u) \vdash \exists y.\, \mathsf{ls}(x, y)$

Proof tree of E_2, using the induction hypothesis **E**

$$
\dfrac{(E_1) \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\mathsf{emp} \vdash \exists y, w.\, (\mathsf{emp} \wedge u{=}w \wedge t{=}y)}\ \text{SMT solvers}}{\mathsf{ls}(u, t) \vdash \exists y, w.\, (\mathsf{ls}(w, y) \wedge u{=}w)}\ \text{Match } \mathsf{ls}(u,t) \text{ and } \mathsf{ls}(w,y)}{x{\mapsto}u * \mathsf{ls}(u, t) \vdash \exists y, w.\, (x{\mapsto}w * \mathsf{ls}(w, y))}\ \text{Match } x{\mapsto}u \text{ and } x{\mapsto}w}{(E_{2'})\ x{\mapsto}u * \mathsf{ls}(u, t) \vdash \exists y.\, \mathsf{ls}(x, y)}\ \text{Unfold } \mathsf{ls}(x,y)}{(E_2)\ x{\mapsto}u * \mathsf{tmp}(u) \vdash \exists y.\, \mathsf{ls}(x, y)}\ \text{Apply IH } \mathbf{E} \text{ with } [u/x] \qquad (E_3)}{(\mathbf{E})\ \mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x, y)}
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Inductive step: prove $(E_2)\ x{\mapsto}u * \mathsf{tmp}(u) \vdash \exists y.\, \mathsf{ls}(x, y)$

Proof tree of E_2, using the induction hypothesis **E**

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Inductive step: prove $(E_2)\ x{\mapsto}u * \mathsf{tmp}(u) \vdash \exists y.\, \mathsf{ls}(x, y)$

Proof tree of E_2, using the induction hypothesis **E**

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$$

Inductive step: prove (E_3) $x \mapsto u * u \mapsto v * \mathsf{tmp}(v) \vdash \exists y.\, \mathsf{ls}(x, y)$

(base step)	(inductive step)	(inductive step)
(E_1)	(E_2)	(E_3)
$\mathsf{emp} \vdash \exists y.\, \mathsf{ls}(x, y)$	$x \mapsto u * \mathsf{tmp}(u) \vdash \exists y.\, \mathsf{ls}(x, y)$	$x \mapsto u * u \mapsto v * \mathsf{tmp}(v) \vdash \exists y.\, \mathsf{ls}(x, y)$

$$\frac{\mathsf{emp} \vdash \exists y.\, \mathsf{ls}(x, y) \qquad x \mapsto u * \mathsf{tmp}(u) \vdash \exists y.\, \mathsf{ls}(x, y) \qquad x \mapsto u * u \mapsto v * \mathsf{tmp}(v) \vdash \exists y.\, \mathsf{ls}(x, y)}{(\mathbf{E})\ \mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x, y)}$$

Inductive step: prove $(E_3)\ x{\mapsto}u * u{\mapsto}v * \mathsf{tmp}(v) \vdash \exists y.\, \mathsf{ls}(x, y)$

Similarly, proof tree of E_3, using induction hypothesis **E**

$$
\dfrac{(E_1) \qquad (E_2) \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\mathsf{emp} \vdash \exists y, z, w.\, (\mathsf{emp} \wedge u{=}z \wedge v{=}w \wedge t{=}y)}\ \text{SMT solvers}}{\mathsf{ls}(v, t) \vdash \exists y, z, w.\, (\mathsf{ls}(w, y) \wedge u{=}z \wedge v{=}w)}\ \text{Match } \mathsf{ls}(v,t) \text{ \& } \mathsf{ls}(w,y)}{u{\mapsto}v * \mathsf{ls}(v, t) \vdash \exists y, z, w.\, (z{\mapsto}w * \mathsf{ls}(w, y) \wedge u{=}z)}\ \text{Match } u{\mapsto}v \text{ \& } z{\mapsto}w}{u{\mapsto}v * \mathsf{ls}(v, t) \vdash \exists y, z.\, (\mathsf{ls}(z, y) \wedge u{=}z)}\ \text{Unfold } \mathsf{ls}(z,y)}{x{\mapsto}u * u{\mapsto}v * \mathsf{ls}(v, t) \vdash \exists y, z.\, (x{\mapsto}z * \mathsf{ls}(z, y))}\ \text{Match } x{\mapsto}u \text{ and } x{\mapsto}z}{x{\mapsto}u * u{\mapsto}v * \mathsf{ls}(v, t) \vdash \exists y.\, \mathsf{ls}(x, y)}\ \text{Unfold } \mathsf{ls}(x,y)}{(E_3)\ x{\mapsto}u * u{\mapsto}v * \mathsf{tmp}(v) \vdash \exists y.\, \mathsf{ls}(x, y)}\ \text{Apply IH } \mathbf{E}}{(\mathbf{E})\ \mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x, y)}
$$

Better strategy: prove E_3 by reusing proof of E_2

Once proof of E_2 is found, we can collect entailments derived in the proof

$$(E_{2'})\ x{\mapsto}u * \mathsf{ls}(u,t) \vdash \exists y.\, \mathsf{ls}(x,y)$$
$$(E_2)\ x{\mapsto}u * \mathsf{tmp}(u) \vdash \exists y.\, \mathsf{ls}(x,y)$$

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And use them as hypotheses to prove E_3

$$\dfrac{(E_1) \qquad (E_2) \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\mathsf{emp} \vdash \exists y.(\mathsf{emp} \wedge y{=}z)}\ \text{SMT solvers}}{\mathsf{ls}(x,z) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Match } \mathsf{ls}(x,z), \mathsf{ls}(x,y).}{x{\mapsto}u * \mathsf{ls}(u,r) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Apply HP } E_{2'}}{(E_3)\ x{\mapsto}u * u{\mapsto}v * \mathsf{tmp}(v) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Apply HP } E_2}{(\mathbf{E})\ \mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x,y)}$$

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Better strategy: prove E_3 by reusing proof of E_2

Once proof of E_2 is found, we can collect entailments derived in the proof

$$(E_{2'})\ x{\mapsto}u * \mathsf{ls}(u,t) \vdash \exists y.\, \mathsf{ls}(x,y)$$
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Better strategy: prove E_3 by reusing proof of E_2

Once proof of E_2 is found, we can collect entailments derived in the proof

$$(E_{2'})\ x{\mapsto}u * \mathsf{ls}(u,t) \vdash \exists y.\, \mathsf{ls}(x,y)$$
$$(E_2)\ x{\mapsto}u * \mathsf{tmp}(u) \vdash \exists y.\, \mathsf{ls}(x,y)$$

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And use them as hypotheses to prove E_3

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Comparison: the new proof and the standard proof of E_3

$$
\dfrac{(E_1) \qquad (E_2) \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\mathsf{emp} \vdash \exists y.(\mathsf{emp} \wedge y{=}z)}\ \text{SMT solvers}}{\mathsf{ls}(x,z) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Match } \mathsf{ls}(x,z), \mathsf{ls}(x,y).}{x{\mapsto}u * \mathsf{ls}(u,r) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Apply HP } E_{2'}}{(E_3)\ x{\mapsto}u * u{\mapsto}v * \mathsf{tmp}(v) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Apply HP } E_2}{(\mathbf{E})\ \mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x,y)}
$$

$$
\dfrac{(E_1) \qquad (E_2) \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\mathsf{emp} \vdash \exists y,z,w.\,(\mathsf{emp} \wedge u{=}z \wedge v{=}w \wedge t{=}y)}\ \text{SMT solvers}}{\mathsf{ls}(v,t) \vdash \exists y,z,w.\,(\mathsf{ls}(w,y) \wedge u{=}z \wedge v{=}w)}\ \text{Match } \mathsf{ls}(v,t)\ \&\ \mathsf{ls}(w,y)}{u{\mapsto}v * \mathsf{ls}(v,t) \vdash \exists y,z,w.\,(z{\mapsto}w * \mathsf{ls}(w,y) \wedge u{=}z)}\ \text{Match } u{\mapsto}v\ \&\ z{\mapsto}w}{u{\mapsto}v * \mathsf{ls}(v,t) \vdash \exists y,z.\,(\mathsf{ls}(z,y) \wedge u{=}z)}\ \text{Unfold } \mathsf{ls}(z,y)}{x{\mapsto}u * u{\mapsto}v * \mathsf{ls}(v,t) \vdash \exists y,z.\,(x{\mapsto}z * \mathsf{ls}(z,y))}\ \text{Match } x{\mapsto}u \text{ and } x{\mapsto}z}{x{\mapsto}u * u{\mapsto}v * \mathsf{ls}(v,t) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Unfold } \mathsf{ls}(x,y)}{(E_3)\ x{\mapsto}u * u{\mapsto}v * \mathsf{tmp}(v) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Apply IH } \mathbf{E}}{(\mathbf{E})\ \mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x,y)}
$$

What are the differences?

Proof theory: standard cut rule

$$\frac{F \vdash K \qquad K \vdash G}{F \vdash G}$$

Induction proof: apply induction hypothesis by using the **cut** rule

$$\frac{\langle A \vdash B \rangle_{IH} \qquad F \vdash A * K \qquad B * K \vdash G}{F \vdash G}$$

Source of cut points:

- SL induction prover [Chu et al., 2015]: induction hypotheses
- **Our work: induction hypotheses + derived entailments**

How about soundness?

$$
\dfrac{(E_1) \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\mathsf{emp} \vdash \exists y, w.\,(\mathsf{emp} \wedge u{=}w \wedge t{=}y)}\ \text{SMT solvers}}{\mathsf{ls}(u,t) \vdash \exists y, w.\,(\mathsf{ls}(w,y) \wedge u{=}w)}\ \text{Match } \mathsf{ls}(u,t) \text{ and } \mathsf{ls}(w,y)}{x{\mapsto}u * \mathsf{ls}(u,t) \vdash \exists y, w.\,(x{\mapsto}w * \mathsf{ls}(w,y))}\ \text{Match } x{\mapsto}u \text{ and } x{\mapsto}w}{(E_{2'})\ x{\mapsto}u * \mathsf{ls}(u,t) \vdash \exists y.\,\mathsf{ls}(x,y)}\ \text{Unfold } \mathsf{ls}(x,y)}{(E_2)\ x{\mapsto}u * \mathsf{tmp}(u) \vdash \exists y.\,\mathsf{ls}(x,y)}\ \text{Apply IH } \mathbf{E} \text{ with } [u/x] \qquad (E_3)}{(\mathbf{E})\ \mathsf{tmp}(x) \vdash \exists y.\,\mathsf{ls}(x,y)}
$$

When finish proving E_2, we don't know about E_3 and **E**!

- $(E_2)\ x{\mapsto}u * \mathsf{tmp}(u) \vdash \exists y.\,\mathsf{ls}(x,y) \rightsquigarrow$ validity: **?**
- $(E_{2'})\ x{\mapsto}u * \mathsf{ls}(u,t) \vdash \exists y.\,\mathsf{ls}(x,y) \rightsquigarrow$ validity: ✓

How about soundness?

Using $E_2(?)$ as a hypothesis to prove E_3

$$\dfrac{(E_1) \qquad (E_2) \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\mathsf{emp} \vdash \exists y.(\mathsf{emp} \wedge y{=}z)}\ \text{SMT solvers}}{\mathsf{ls}(x,z) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Match } \mathsf{ls}(x,z), \mathsf{ls}(x,y).}{x{\mapsto}u * \mathsf{ls}(u,r) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Apply HP } E_{2'}}{(E_3)\ x{\mapsto}u * u{\mapsto}v * \mathsf{tmp}(v) \vdash \exists y.\, \mathsf{ls}(x,y)}\ \text{Apply HP } E_2}{(\mathbf{E})\ \mathsf{tmp}(x) \vdash \exists y.\, \mathsf{ls}(x,y)}$$

How to conclude about E_3 and then $\mathbf{E}$?

$\Rightarrow$ Consider the mutual induction proof of $\mathbf{E}$, E_2, E_3

Mutual induction principle

In order to prove entailments $F_1 \vdash G_1, ..., F_n \vdash G_n$ satisfied by all models s, h (i.e., all $F_1 \vdash G_1, ..., F_n \vdash G_n$ are Valid), it suffices to prove that: $F_1 \vdash G_1$, ..., $F_n \vdash G_n$ hold for all s, h, whenever $F_1 \vdash G_1$, ..., $F_n \vdash G_n$ hold for all $s', h' \prec s, h$

Formally:

$$\frac{\forall s,h.\ (\forall s',h'.\ s',h' \prec s,h \rightarrow s',h' \models \{F_i \vdash G_i \mid i{=}1..n\}) \rightarrow s,h \models \{F_i \vdash G_i \mid i{=}1..n\}}{\forall s,h.\ s,h \models \{F_i \vdash G_i \mid i{=}1..n\}}$$

Proof system: general inference rules

$$(\bot\,\mathsf{L}_1)\ \frac{}{\mathcal{H},\ \rho,\ F_1 \wedge u{\neq}u \vdash F_2}$$

$$(\bot\,\mathsf{L}_2)\ \frac{}{\mathcal{H},\ \rho,\ F_1 * u \overset{\iota_1}{\mapsto} \vec{v} * u \overset{\iota_2}{\mapsto} \vec{w} \vdash F_2}$$

$$(\vdash_{\mathsf{pure}})\ \frac{}{\mathcal{H},\ \rho,\ \Pi_1 \vdash \Pi_2}\ \Pi_1 \Rightarrow \Pi_2$$

$$(\mathsf{empL})\ \frac{\mathcal{H},\ \rho',\ F_1 \vdash F_2}{\mathcal{H},\ \rho,\ F_1 * \mathsf{emp} \vdash F_2}$$

$$(=\mathsf{L})\ \frac{\mathcal{H},\ \rho',\ F_1[u/v] \vdash F_2[u/v]}{\mathcal{H},\ \rho,\ F_1 \wedge u{=}v \vdash F_2}$$

$$(\mathsf{empR})\ \frac{\mathcal{H},\ \rho',\ F_1 \vdash \exists \vec{x}.F_2}{\mathcal{H},\ \rho,\ F_1 \vdash \exists \vec{x}.(F_2 * \mathsf{emp})}$$

$$(=\mathsf{R})\ \frac{\mathcal{H},\ \rho',\ F_1 \vdash \exists \vec{x}.F_2}{\mathcal{H},\ \rho,\ F_1 \vdash \exists \vec{x}.(F_2 \wedge u{=}u)}$$

$$(*\mapsto)\ \frac{\mathcal{H},\ \rho',\ F_1 \vdash \exists \vec{x}.(F_2 \wedge u{=}t \wedge \vec{v}{=}\vec{w})}{\mathcal{H},\ \rho,\ F_1 * u \overset{\iota}{\mapsto} \vec{v} \vdash \exists \vec{x}.(F_2 * t \overset{\iota}{\mapsto} \vec{w})}\ (u, \vec{v}) \# \vec{x}$$

$$(\exists\,\mathsf{L})\ \frac{\mathcal{H},\ \rho',\ F_1[u/x] \vdash F_2}{\mathcal{H},\ \rho,\ \exists x.F_1 \vdash F_2}\ u \notin FV(F_2)$$

$$(*\,\mathsf{P})\ \frac{\mathcal{H},\ \rho',\ F_1 \vdash \exists \vec{x}.(F_2 \wedge \vec{u}{=}\vec{v})}{\mathcal{H},\ \rho,\ F_1 * \mathsf{P}(\vec{u}) \vdash \exists \vec{x}.(F_2 * \mathsf{P}(\vec{v}))}\ \vec{u} \# \vec{x}$$

$$(\exists\,\mathsf{R})\ \frac{\mathcal{H},\ \rho',\ F_1 \vdash F_2[e/x]}{\mathcal{H},\ \rho,\ F_1 \vdash \exists x.F_2}$$

$$(\mathsf{PR})\ \frac{\mathcal{H},\ \rho',\ F_1 \vdash \exists \vec{x}.(F_2 * F_i^{\mathsf{P}}(\vec{u}))}{\mathcal{H},\ \rho,\ F_1 \vdash \exists \vec{x}.(F_2 * \mathsf{P}(\vec{u}))}\ F_i^{\mathsf{P}}(\vec{u}) \Rightarrow \mathsf{P}(\vec{u})$$

$\mathcal{H}$: set of hypotheses, ρ : a trace of used rules in a proof path

Proof system: induction rules

$$(\text{Ind})\ \frac{\mathcal{H} \cup \{(H, \mathbf{?})\},\ \rho',\ F_1 * F_1^{\mathsf{P}}(\vec{u}) \vdash F_2 \quad \ldots \quad \mathcal{H} \cup \{(H, \mathbf{?})\},\ \rho',\ F_1 * F_m^{\mathsf{P}}(\vec{u}) \vdash F_2}{\mathcal{H},\ \rho,\ F_1 * \mathsf{P}(\vec{u}) \vdash F_2}\ \dagger_{(\text{Ind})}$$

with $H \triangleq F_1 * \mathsf{P}(\vec{u}) \vdash F_2$, $\rho' = (\text{Ind}) :: \rho$,
and $\dagger_{(\text{Ind})}$: $\mathsf{P}(\vec{u}) \triangleq F_1^{\mathsf{P}}(\vec{u}) \vee \ldots \vee F_m^{\mathsf{P}}(\vec{u})$

Proof system: induction rules

$$(\mathsf{Ind})\ \frac{\mathcal{H} \cup \{(H, ?)\},\ \rho',\ F_1 * F_1^{\mathsf{P}}(\vec{u}) \vdash F_2 \quad \ldots \quad \mathcal{H} \cup \{(H, ?)\},\ \rho',\ F_1 * F_m^{\mathsf{P}}(\vec{u}) \vdash F_2}{\mathcal{H},\ \rho,\ F_1 * \mathsf{P}(\vec{u}) \vdash F_2}\ \dagger_{(\mathsf{Ind})}$$

with $H \triangleq F_1 * \mathsf{P}(\vec{u}) \vdash F_2$, $\rho' = (\mathsf{Ind}) :: \rho$,
and $\dagger_{(\mathsf{Ind})}$: $\mathsf{P}(\vec{u}) \triangleq F_1^{\mathsf{P}}(\vec{u}) \vee \ldots \vee F_m^{\mathsf{P}}(\vec{u})$

$$(\mathsf{AH})\ \frac{\mathcal{H} \cup \{(H, \mathit{status})\},\ (\mathsf{AH}) :: \rho,\ F_4\theta * \Sigma' \wedge \Pi_1 \vdash F_2}{\mathcal{H} \cup \{(H \triangleq \Sigma_3 \wedge \Pi_3 \vdash F_4, \mathit{status})\},\ \rho,\ \Sigma_1 \wedge \Pi_1 \vdash F_2}\ \begin{array}{l}\exists\theta, \Sigma'.(\Sigma_1 \cong \Sigma_3\theta * \Sigma' \wedge \Pi_1 \Rightarrow \Pi_3\theta), \\ \dagger_{(\mathsf{AH})}\end{array}$$

with $\dagger_{(\mathsf{AH})}$: $(\mathit{status} = \checkmark)$
$\vee\ \exists \iota, u, \vec{v}, \Sigma''.(\Sigma' \cong u \overset{\iota}{\mapsto} \vec{v} * \Sigma'')$
$\vee\ \exists \rho_1, \rho_2.(\rho = \rho_1 @ [(* \mapsto)] @ \rho_2 \wedge (\mathsf{Ind}) \notin \rho_1 \wedge (\mathsf{Ind}) \in \rho_2)$.

Proof system: search procedure

Procedure Prove($\mathcal{H}, \rho, F \vdash G$)

Input: Set of hypotheses, proof trace and the goal entailment.

Output: Derivation result, a set of derived entailments, and applied hypotheses.

$\mathcal{S} \leftarrow \{R_{inst} \mid R_{inst} = \text{Unify}(R, (\mathcal{H}, \rho, F \vdash G)) \ \wedge \ R \in \mathcal{R}\}$

if $\mathcal{S} = \varnothing$ **then return** False, $\varnothing, \varnothing$ *// no rule is selected*

for each R_{inst} **in** $\mathcal{S}$ **do**

  if GetName(R_{inst}) $\in \{\vdash_{\text{pure}}, \bot\,\text{L}_1, \bot\,\text{L}_2\}$ **then** *// R is an axiom rule*

    return True, $\varnothing, \varnothing$

  $\mathcal{H}_{used} \leftarrow \varnothing$

  if R_{inst} = AH with hypothesis **E then** $\mathcal{H}_{used} \leftarrow \mathcal{H}_{used} \cup \{E\}$

  $\mathcal{H}_{derived} \leftarrow \varnothing$

  $(\mathcal{H}_i, \rho_i, F_i \vdash G_i)_{i=1,\ldots,n} \leftarrow$ GetPremises(R_{inst}) *// all premises of R_{inst}*

  for i = 1 **to** n **do**

    $res, \mathcal{H}_{derived}, \mathcal{H}'_{used} \leftarrow$ Prove($\mathcal{H}_i \oplus \mathcal{H}_{derived}, \rho_i, F_i \vdash G_i$)

    if res = False **then return** False, $\varnothing, \varnothing$

    $\mathcal{H}_{used} \leftarrow \mathcal{H}_{used} \cup \mathcal{H}'_{used}$

  if $\mathcal{H}_{used} \subseteq$ (GetEntailments($\mathcal{H}_{derived}$) $\cup \{F \vdash G\}$) **then**

    $\mathcal{H}_{derived} \leftarrow \mathcal{H}_{derived} \oplus \{(F \vdash G, \checkmark)\}$

  else $\mathcal{H}_{derived} \leftarrow \mathcal{H}_{derived} \oplus \{(F \vdash G, ?)\}$

  return True, $\mathcal{H}_{derived}, \mathcal{H}_{used}$ *// all derived premises are proved*

return False, $\varnothing, \varnothing$ *// all rules fail to prove $F \vdash G$*

Experiment

Implementation:

- The separation logic induction prover: **Songbird**
- http://loris-5.d2.comp.nus.edu.sg/songbird/

Experiment:

- Benchmarks: *slrd_entl* from SL-COMP 14, and *slrd_indt* for induction provers
- Testing environment: Ubuntu server 14.04, CPU Intel E5-2620 (2.4GHz), RAM 64GB.

Experiment: slrd_entl benchmark from SL-COMP 2014

Category	Slide	Spen	Sleek	Cyclist	Songbird
singly-ll (64)	12	3	48	**63**	**63**
doubly-ll (37)	14	0	17	24	**26**
nested-ll (11)	0	**11**	5	6	**11**
skip-list (13)	0	**12**	4	5	7
tree (26)	12	1	14	18	**22**
Total (151)	38	27	88	116	**129**

Overall comparison on number of proved entailments (T/O: 30s)

Slide: [Iosif et al., 2013]
Spen: [Enea et al., 2014]
Sleek: [Chin et al., 2012]
Cyclist: [Brotherston et al., 2011]

	Songbird			
	$\checkmark_{sb}\,\times_{o}$	$\times_{sb}\,\checkmark_{o}$	$\checkmark_{sb}\,\checkmark_{o}$	$\times_{sb}\,\times_{o}$
Cyclist	13	0	116	22
Sleek	41	0	88	22
Spen	109	7	20	15
Slide	103	12	26	10

Detailed comparison

Experiment: slrd_indt benchmark for induction provers

Category		Cyclist	$\textbf{Songbird}_{\textbf{SI}}$	Songbird
ll/ll2	(24)	18	22	**24**
ll-even/odd	(20)	8	17	**20**
ll-left/right	(20)	12	10	**20**
misc.	(32)	17	16	**32**
Total	(96)	55	65	**96**

Number of proved entailments (T/O: 30s)

Note: $\textbf{Songbird}_{\textbf{SI}}$ - a variant of **Songbird**, whose mutual induction feature is disable, i.e., supporting only <u>s</u>tandard <u>i</u>nduction proof.

Conclusion

Theory:

- Mutual induction proof for separation logic entailments.
- Detailed proof system: inference rules and a search procedure.

Practice:

- The automated induction prover **Songbird**:
  http://loris-5.d2.comp.nus.edu.sg/songbird/
- Experiment with 2 benchmarks slrd_entl (SL-COMP 14) and slrd_indt (for induction provers)

Future works

- Support bi-abduction reasoning
  $$F \quad * \quad ?AntiFrame \ \vdash \ G \quad * \quad ?Frame$$ [1]
- Integrate into a software verification system [2] to verify imperative programs (C/C++/Java)
- Handle more separation logic fragments: separating implication ($-\!\!*$) with user-defined inductive heap predicates

[1] Infer : http://fbinfer.com/docs/separation-logic-and-bi-abduction.html
[2] HIP/SLEEK: http://loris-5.d2.comp.nus.edu.sg/hip/

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– *Andrew W. Appel et al.: Program Logics for Certified Compilers, 2014*

Thank you!

Thank you!

Take-away messages:

- Mutual induction proof for separation logic entailments
- The automated separation logic induction prover **Songbird**:
  http://loris-5.d2.comp.nus.edu.sg/songbird/